

|                 |                                                                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b>        |                                                                                                                                                                                      |
| <b>(a)</b>      | Weight of cylinder = $Ah\rho g$                                                                                                                                                      |
|                 | Pressure = $\frac{F}{A} = \frac{Ah\rho g}{A}$                                                                                                                                        |
| <b>(b)(i)</b>   | An ideal gas is one where there are no intermolecular forces.                                                                                                                        |
|                 | An ideal gas obeys the ideal gas law $pV = nRT$<br>where $p$ = pressure,<br>$V$ = volume, $n$ = number of moles, ( $R$ = molar gas constant) and<br>$T$ = thermodynamics temperature |
| <b>(b)(ii)</b>  | Since it is in <u>equilibrium</u> , the pressure of the trapped gas is the sum of the atmospheric pressure and the pressure due to the weight of the mercury.                        |
|                 | $(p_o + p_m) \times 0.190 = \frac{nRT}{A}$                                                                                                                                           |
| <b>(b)(iii)</b> | Recognize that $\frac{nRT}{A}$ is a constant OR $190 p_o + 190 p_m = 208 p_o - 208 p_m$                                                                                              |
|                 | $p_m = 0.035 \times 1.36 \times 10^4 \times 9.81 = 4.67 \times 10^3 \text{ Pa}$<br>$p_o = 1.03 \times 10^5 \text{ Pa}$                                                               |
| <b>(b)(iv)</b>  | Since temperature stay constant, there is no change to the internal energy of the gas.                                                                                               |
|                 | Negative work done results in a postive Q since $\Delta U=0$ .                                                                                                                       |

Anglo-Chinese Junior College  
H2 (9749) Physics

---

2021 H2 Preliminary Exam Paper 3 MS